Multimodal Glioma Segmentation and Classification and NLP Analysis of Scholarly Articles to investigate how to improve Segmentation.

Technical Document

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Problem Statement:

All Neuro-oncologists should be able to seamlessly and accurately determine complete regions of the brain which constitute gliomas and identify the glioma as high grade or low grade without performing invasive and stressful methods towards the patient.

Introduction:

A glioma is the presence of a common tumor in glial cells of the brain where approximately 33 percent of all brain tumors are gliomas. The malignant glial cells that supports neurons in brain are comprised of astrocytes, oligodendrocytes and ependymal cells. These cells are key components in the central nervous system responsible for a variety of processing and structural activities vital for neurological proclivity. Astrocytes are essential for synaptic support and axon guidance within the Central Nervous System (CNS) and helps to regulate the control of the blood brain barrier and blood flow. Oligodendrocytes are responsible for the insulation of axons and are the end products of a cell lineage that are exposed to proliferation, migration, differentiation and myelination. Ependymal cells possess and array of important functions in a developing brain, however are no longer needed in mature brains. They are currently best known for the transport of nutrients and electrolytes between the cerebrospinal fluid and the brain parenchyma.

The detection, segmentation and classification of brain tumors are particularly important in medical industry as it is imperative to provide the necessary diagnosis to patients without performing invasive methods that improves the timeliness in which treatment can be administered. The current armamentarium in modern medicine is still unable to provide any known cure for gliomas to the extent where all low grade gliomas will progress to high grade gliomas (Glioblastoma Multiform). As such early detection is paramount in improving survival rates where contemporary treatment methods may inhibit the progression of glioma tumors.

To achieve seamless classification and segmentation of gliomas an automated approach would be the most optimal as opposed to expert analysis on individual MRI scans which would be time consuming. Automation involves the building of models that are independent of human intervention for classification and segmentation purposes, however is highly dependent on machine learning principles responsible for how models would be obtained. One of the most effective ways of building image segmentation and classification models is through convolutional neural networks especially in the medical industry. This is mainly due to its high accuracy output produced and this is supported by the fact that images are subjected to numerous layers within the network which are all connected by weights or neurons through a hierarchical model. The relationship between layers perform feature parameter sharing as well as dimensionality reduction reducing computational expense but increasing accuracy. Neuro experts or neuro-oncologists have been investigating key features of gliomas and documented their finding through scholarly articles. As a result it is important to consider salient arguments from these documents related to glioma features in order to validate whether or not datasets that will be used for training are adequately prepared and or processed before loading within a convolutional neural network required for model preparation.

This research paper seeks to build a tool that will be able to detect, segment and classify gliomas by uploading images. The basis to which this solution will be achieved is by generating independent models from a convolutional neural network (CNN) for segmentation and classification. These models should be accessible via a user interface that will be able to analyze images of the brain from any plane within the dimensions of sagittal, coronal and axial. In addition various types of weighted images (t1, t2, flair, t1ce) will be accounted for within the models for both classification and segmentation. Additional investigation of scholarly articles will be done to verify if the features labelled within datasets are sufficient for the purpose of image segmentation.

Data Description:

<u>Glioma Scholarly Articles</u>

This dataset consist of 92 different scholarly articles related to glioma, necrosis, and edema and enhancing tumor between years 2010 – 2020. The articles were randomly downloaded via pypaperbot. Total disk size is 187 MB

<u>Glioma Dataset</u>

Source for this dataset is located at https://www.kaggle.com/aryashah2k/brain-tumor-segmentation-brats-2019. Total data size is 25.74 GB and the format of images are nii extensions. Each nii file is weighted as T1, T2, T1CE and FLAIR and each contain planes of the brain for dimensions sagittal, coronal and axial consisting of 155 slices each. For each slice of the brain there is an associated segment that corresponds to each weight (T1, T2, T1CE and FLAIR) and plane (sagittal, axial and coronal). These scans are regarded as multimodal since they cover different planes and weights and are collected via different clinical protocols and various scanners and institutions. The segmentation for each slice were all done manually by up to four raters or experts. Labelling of the segments took into account enhancing tumor, edema and necrotic core. Finally the complete dataset separate nii files in distinct folder based on whether or not the glioma is a high grade or low grade. Therefore all nii files for High grade gliomas were in a separate folder called HGG and all nii files associated with low grade gliomas were in another folder called LGG.

Note: Survival data from this dataset was not used for this paper.

Procedures (All procedures carried out using a Jupyter notebook / google colab):

Naturual Language Processing

Tool: Pypaperbot was installed via “pip install pypaperbot”

The following script was run from command prompt:

python -m PyPaperBot --query="glioblastoma" --scholar-pages=11 --min-year=2010 --dwn-dir="path" --scihub-mirror=https://sci-hub.do

This downloaded up to 10 papers that included the query “glioblastoma” that was published starting from year 2010. The script was done repeatedly using other key words as queries such as Astrocytomas, Ependymomas, Glial Tumors, Glioblastoma, Glioblastoma Multiforme, Glioma, High Grade Glioma, Low Grade Glioma, Oligodendrogliomas and Pilocytic Astrocytoma.

NLP analysis was done on each paper in attempt to find the significance of expert labels as well as to identify if there are any other feature of glioma that could have been included in segmentation process.

NLP Process regarding key word extraction:

1) Scholarly articles were downloaded using pypaperbot
2) Duplicate documents downloaded from each command line run was removed
3) All non-duplicates were placed in a consolidated folder
4) Data paths were loaded to a data frame based on documents that exist in that folder.
5) The text of each document were read using parser and appended to the data frame where the content matches the data path and document title.
6) Texts were encoded as utf-8 and unwanted strings were replaced with appropriate placeholders such as “\n” replaced with “”.
7) Texts were split by spaces (“ “).
8) Most common words in each article were identified
9) Least common word in each article were identified
10) wordnet, stopwords, and words were downloaded from nltk
11) Stopwords, Porterstemmer and RegexpTokenizer and WordNetLemmatizer were imported
12) Total stopwords were set as English words
13) Manual stop words were assigned as newwords which were not relevant to key features such as adobe, shown and organization etc.
14) Stop words and manual stop words were combined to create a complete set of words that should be excluded from key words extraction.
15) Keywords were extracted by removal of punctuation within the text for each document
16) Each word was converted to lower case
17) Tags and special characters were removed

18) A column (keywords) was created on the dataframe where a string of keywords for each document was assigned.
19) An additional column (english_words) was created only extracting English keywords from the keywords column.
20) Create a wordcloud image to view main words from English words list / English_words column).
21) Image accounted for stopwords, random state was equal to 42 and only displayed a maximum of 100 words
22) Verified that minimal outliers existed (pdf, adobe, metatdata, acrobat, window etc.). If any outliers existed additional updates were made to stopwords list.
23) Unigrams, Bigrams and Trigrams were extracted from English_words column.
    a. This was achieved by fitting the English word string via count vectorizer (In order for text to be interpreted as machine learning algorithms, tokenization and vectorisation must be done)
    b. Converting that into bag of words
    c. Take a sum of the words
    d. Calculate the frequency of the word
    e. Sort frequency as highest to lowest
    f. Display results as a dataframe and bar chart for the top 20 frequent words.
    g. This procedure was repeated for bigrams and trigrams however for step "a" range (2,2) was used for bigrams and range (3,3) was used for trigrams, as well as a maximum of 2000 features were utilized.
    h. Each dataframe for unigram, bigram and trigram were combined and saved a csv for reporting on a dashboard.
24) The english_word column was refined to a matrix of integers using tf-idf vectoriser. This will allow the extraction of context specific words. Therefore only the most important word in context was obtained. TF is defined as term frequency and IDF is defined as Inverse document frequency. Individual formulas given below:
    a. TF = Frequency of a term in a document / total number of terms in the document
    b. IDF = log(Total documents) / # of documents with the term
    c. Analysis done to verify if any key feature from context words could have been used for segmentation
25) The total number of times that necrosis, edema and enhancing tumor was used in each document was calculated and the significance was measured.
26) The total amount of documents that mentioned necrosis, edema and enhancing tumor was also identified.
    a. Words equated to necrosis: Necrotic Core, Necrotic, Necrosis
    b. Words equated to Edema: Oedema, Edema, Swelling
    c. Words equated to Enhancing Tumor: Enhanced Tumor, Enhancing glioma, Enhancing Tumor
    d. If neither of the three experts words were used then label assigned is None

27) The total amount of main words also calculated but was not used for analysis since they are not context words.

Image Processing

Preparation for image segmentation

1) A data frame was prepared that referenced the data path for each nii image which included t1, t2, t1ce and flair and their associated segment within the same column.
2) A function save fig was created to read nii files and the features of this function included:
   a. Ability to read all planes (sagittal, coronal, and axial) of nii files.
   b. Ability to extract all 155 slices from each image type (t1, t2, t1ce, flair and seg)
   c. Normalize images within range 0 to 255
   d. Convert images to uint8 format
   e. Renaming images using format gliomatype, image plane, existing bratslabel, slice number and file extension as png
3) A “for” loop was used to reference each nii file path in the dataframe and the save fig function was used to extract individual images and saved them to a designated folder.
4) Images were resized to 256 by 256 in size

Preparation for Image Classification

1) A dataframe for all images was created (consisting of only one column)
2) Images that were segments and not brain slices were separated from that dataframe as an independent dataframe
3) A subfolder column was created for each dataframe based on brats labels or annotations in addition to image slice number
4) A merge was performed between dataframes using brats label and slice number so that each image slice would be aligned with the appropriate segment slice.
5) Created a column that calculated the pixel size of each image and segment
6) Created a column that calculated the sharpness or contrast of images
7) Select only the fraction of dataframe where image and segment pixel count is greater than the mean and image contrast was greater than the mean.
8) Created a column for Tumor Class to verify if image is a high grade, lowgrade or does not have a tumor. The process outlined below:
   a. Used getbbox function to check if the segment from the dataframe contains a image
   b. If the segment if blank or empty then that image does not possess a tumor
   c. Otherwise if the image name starts with “HGG” it’s a high grade glioma else a low grade glioma

   d. 85% of the dataframe rows were extracted as training images and the other 15% as testing images. Of these amounts only 25% were extracted from training and testing.
9) The extracted images were stored in separate folders as HGG, LGG or NO_TUMOR.

Segmentation CNN Model:

1) A dataframe with image paths and their associated segmented path was prepared
2) Images and segments were resized to and 240 x 240 dimensions
3) Dataframe for images were split into train and test set 75% for training 25% for testing
4) Epochs, batch size, Imgheight, Imgwidth and Channels were specified
5) Data Generator and augmentation was configured which included rotation_range as 0.2, width_shift_range as 0.05, Height_shift_range as 0.05, shear_range as 0.05, zoom_range as 0.05, horizontal_flip as True, fill_mode as 'nearest'.
6) Training Data consisted of using the Image paths from the training dataframe to create training images and segment paths were used for labelling.
7) Test or Validation data consisted of using the Image paths from the testing dataframe to create training images and segment paths were used for labelling.
8) A unet convolutional neural network was constructed for image segmentation.
   a. Convolutional neural network initially constructed with 10 layers on a contracting path (Each Layer is accompanied by Max Pooling and Droupout to manage and minimize overfitting). Contracting path begans with 240 x 240 x 1 input size and contracts to 15 x 15 x 256 at the last layer.
   b. On Expansive Path Convolutional neural network was constructed with 4 convolutional transpose layers responsible for both upsampling and and interpretation of coarse data. Expansive path begans with 15 x 15 x 256 input size and expands to 240 x 240 x 1 at the last layer.
      i. Subsequently transpose results are concatenated with convolutional layers from contracting path followed by dropout and 8 additional convolutional layers.
9) Call back parameters were defined for Early Stopping, ReduceLROnPlateau and ModelCheckpoint.
10) Model fit completed considering training data, the amount of steps per epoch, batch size, epochs, callbacks, validation_data and validation steps.
11) Metric used to assess segmentation included dice score, accuracy, specificity and sensitivity.
12) Learning Cureve plotting to graphically assess performance.
13) Model was saved as a .h5 file on local directory
14) Evaluation of model done on data that was not used for training
15) Sample plot completed to show original image, original segment against predicted segment and predicted segment ontop of original image.

Classification CNN Model:

1) Images were stored within respective Training and Testing directories both of which contained subdirectories “HGG”, “LGG” and “NO_TUMOR”, where HGG contained images of High grade gliomas, LGG contained images of Low grade glioma and NO_TUMOR contained images that did not possess a glioma.
2) The labels for each image were there assigned as HGG, LGG and NO_TUMOR
3) All training and testing images were extracted as an array and resized to 240 x 240
4) Training data was shuffled to randomize distribution
5) Training and test data was split 90% to 10% respectively.
6) One hot encoding was used to assign labels to training and test sets
7) Training data was supplemented with pre-trained weights using EfficientNetB0
8) Convolutional Neural Network executed with Transfer Learning comprised of structure:-
    a. Global Average Pooling: acts similar to Max Pooling in CNNs however uses mean values instead of Max values which reduces the need for more compute power.
    b. Dropout: omits neurons at various steps making neurons more independent from the neighboring neurons. Prevents overfitting and neurons are omitted randomly.
    c. Dense: This is the output layer that classifies the image into1 of the 3 possible outcomes. Softmax function is utilized as a generalization of sigmoid function.
9) Model compiled stipulating loss, optimizer and metrics
10) Call back parameters defined for tensorboard to track logs, checkpoints to save model during progression, monitor validation accuracy and reduce_lr to track validation accuracy, patience, mode etc.
11) Model fit done accounting for 12 epochs, a batch size of 32 and relevant callback parameters
12) Graph generated displaying training and validation accuracy and training and validation loss
13) Test prediction done on images not used from training
14) Classification report done to show accuracy of predictions for each label and overall accuracy
15) Model then saved as .h5 file.

Image Segmentation and Classification Tool

1) Models for segmentation and classification were stored in a local directory
2) Parameter for models were defined in relation to Dice Score as (dice_coef), Intersect Over Union (iou), Precision, sensitivity and specificity.
3) Segmentation model was loaded along with custom objects (dice_coef, iou, sensitivity and specificity).
4) Voila was installed for Jupyter notebook to facilitate user interface
5) Ipywidgets was downloaded and imported for building widgets and buttons. Widget List:
    a. Segmentation and classification button- used to segment and classify images
    b. Patient Analysis button – used to segment and classify test images
    c. Upload button – Used to upload images
    d. Upload Patient – Used to upload all images related a patient
    e. Clear button – used to clear segmentation and classification of images
    f. Delete button – used to delete output of patient analysis
    g. Extract load results button – used to extract results of segmentation and classification
    h. Year slider – used to select specific year of scholarly document

i. grams_dropdown – used to select contiguous variables from word frequency using n-grams.

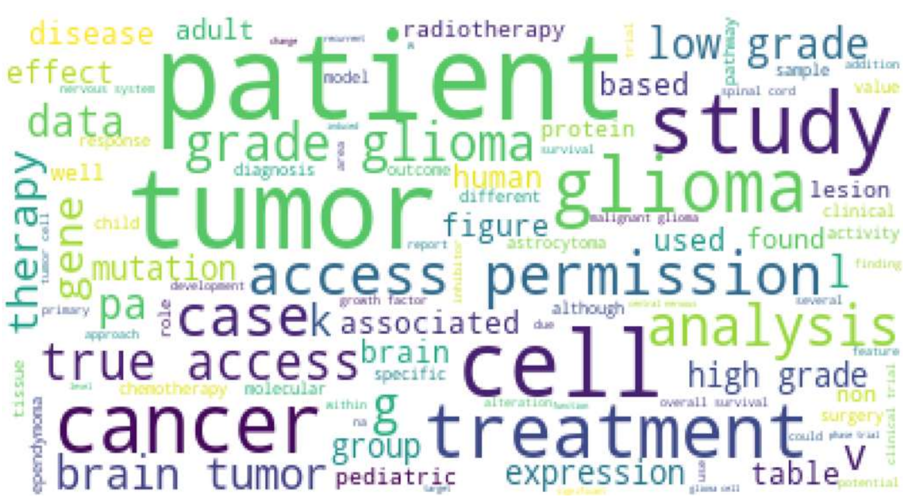
6) A title header was created for dashboard using Jupyter Markdown
7) Four separate tabs were created for dashboard.
   a. Tab 1 contained widgets to analyze patient and was labeled Prediction Analysis’
      i. Widgets included Patient Analysis button, Upload Patient, Delete button
   b. Tab2 contained widgets to perform segmentation and classify individual images and was labelled Prediction Tool.
      i. Widgets included Segmentation and classification button, Upload button, Clear button, Extract load results button
   c. Tab3 contained widgets to view frequency of expert words in documents year over year and was labelled Expert Word Analysis.
      i. Only widget used for tab was a Year slider
   d. Tab4 contained widgets to view n-grams results and was labeled N-Grams
      i. Only widget used for this tab was the n-gram drop down widget

Testing of images provided by UWI.

1) A Jupiter Notebook was created to validate how many images were properly labelled as high grade, low grade glioma and non-tumor.
   a. Note all images provided by the UWI were High grade gliomas.

Observations:

NLP results

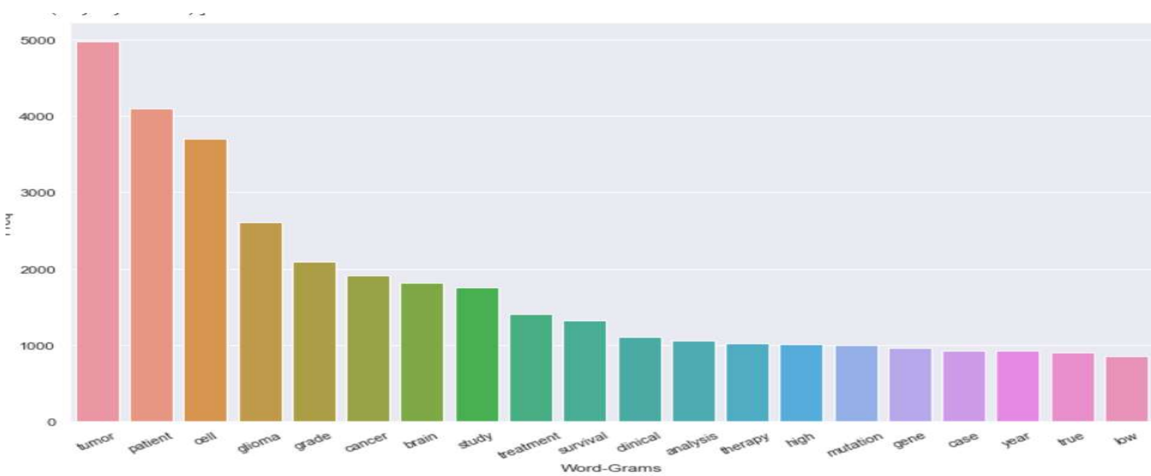


The results obtained from NLP analysis indicated that necrotic core, edema and enhancing tumor although were not found to be highly used contextually across documents still appeared in majority of articles. Words related to Necrotic Core occurred in 48 different documents, words related to edema occurred in 32 different documents and words related to enhancing tumor occurred in 19 different documents. In addition the frequency of of each term Necrotic Core, Edema, Enhancing Tumor across documents was equal to 162, 96 and 62 respectively.

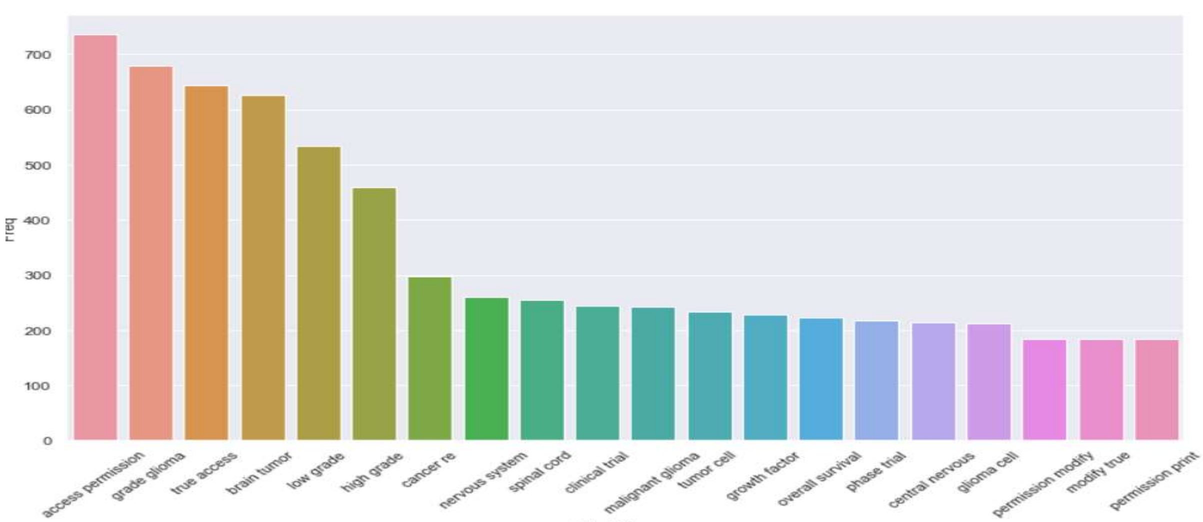
Contiguous Words

The Analysis of contiguous words was accomplished using N-grams where the results of unigrams and bigrams are displayed below.

Unigram



Bigram



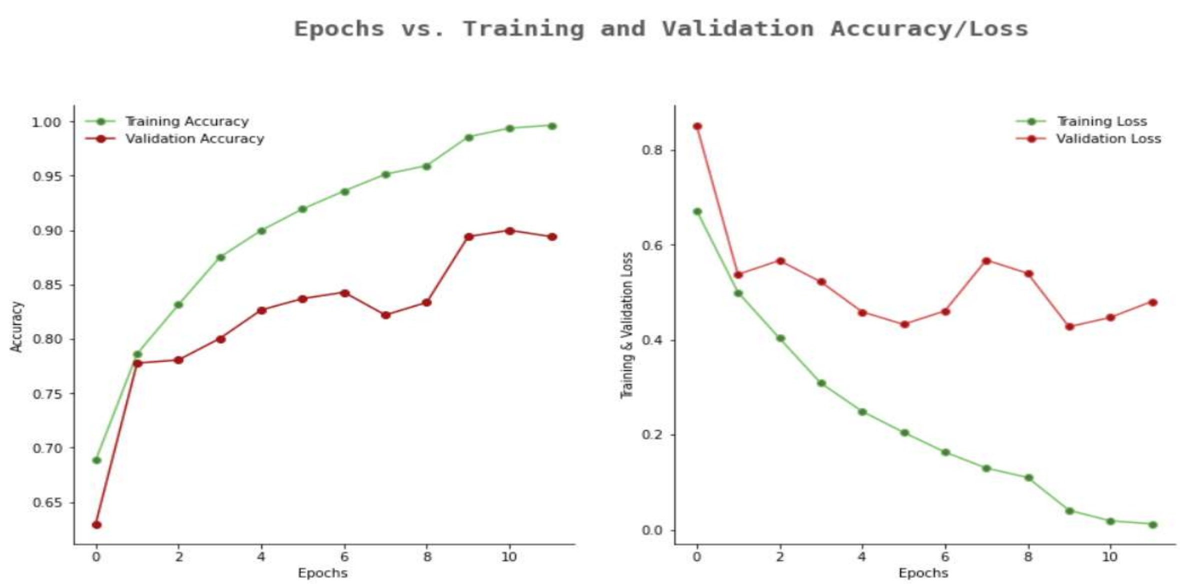
Top 10 Useful contextual words or phrases and their associated score:

Contextual Keywords	Score
Endothelial	0.319
Glioma stem	0.292
Vessel	0.276
Endothelial cell	0.26
Stem	0.248
Glioma stem cell	0.231
Blood Vessel	0.214
Stem cell	0.207
Vascular	0.184
Blood	0.154

<u>CNN Model Performance</u>

Classification

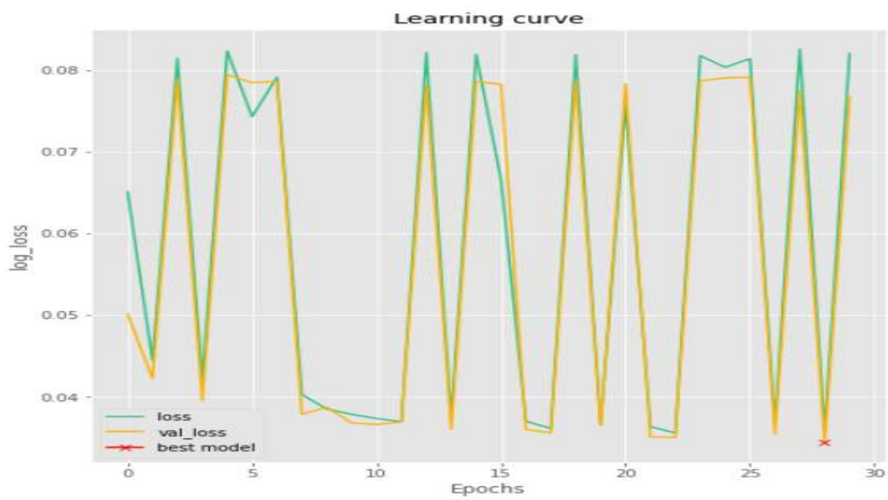
```
              precision    recall  f1-score   support

           0       0.91      0.96      0.93      1086
           1       0.91      0.83      0.87       398
           2       0.88      0.84      0.86       426

    accuracy                           0.90      1910
   macro avg       0.90      0.87      0.89      1910
weighted avg       0.90      0.90      0.90      1910
```



Segmentation

```
#evaluating model on validation data
eval_results = model.evaluate(valid_gen, steps=STEP_SIZE_VALID, verbose=1)
```

```
1491/1491 [==============================] - 71s 47ms/step - loss: 0.0345 - accuracy: 0.9706 - dice_coef: 0.5400 - iou: 0.3703 - precision: 0.4246 - sensitivity: 0.6378 - specificity: 0.9785
```



Tabular Representation for both Segmentation and Classification

Metric	**Classification**	**Segmentation**
Accuracy	99%	97 %
Validation Accuracy	90%	95%
Dice Score	-	54%
Specificity	-	97%
Sensitivity	-	64%
Precision	-	43%

Test Image results

Segmentation was validated against 44 test images containing visible and labelled glioma segments, forty two of which were accurately segmented.

Classification was validated against 44 test images containing high grade gliomas. Thirty Three of which were accurately predicted.

Metric	Segmentation	Classification
Accuracy	95%	75%

Discussion:

NLP: The results obtained for expert words conveys that the choice to produce segments based on enhancing tumor, necrotic core and edema was justified since more frequency of each term spanned more than 90% of all the 10 years in which scholarly articles were downloaded. It is important to note that these terms or phrases did not turn up in the top 10 contextual words suggesting that these phrases were not main focal points from the corpus. It was however observed that from the top 10 words the feature that may aid to improve segmentation analysis would be the inclusion of blood vessel formation (angiogenesis) in the vicinity of gliomas. This has been accounted for in the scholarly article "Angiogenesis in Glioblastoma"where a variety of pathways have been described for the formation of new blood vessels in glioblastoma multiforme. Hypoxic tumor cells, especially those surrounding the necrotic core, release vascular growth factors, such as vascular endothelial growth factor (VEGF), that stimulate the formation of new blood vessels from preexisting normal endothelial cells (Panel A) (Das, Sunit, et al , October 17, 2013). Therefore if segments were somehow modified to include the vascular abnormalities then model may significantly improve. The other contextual phrase to highlight is glioma stem cell which are partly responsible for the differentiation of cells that promote angiogenesis. Although this term may not be feasible to account for in MRI images it fortifies the point that aberrant blood vessel formation is a significant feature to consider.

Segmentation Model:

Segmentation models were evaluated using mainly by dice score followed by accuracy, specificity, sensitivity and intersect over union. The dice score obtained was 60% and on validation data only a 40% score was obtained. This may be due to the further preprocessing steps required and or additional layers of neural network required in order to obtain greater results. The preprocessing steps account for low pixels, low contrast however there were evidence of incomplete images (part of the brain slice was not visible). In addition the images provided for testing included tissues that were not included in the training dataset as a result additional variables were introduced creating noise. The model appears to be unstable and so further preprocessing may be required to improve results

Classification model:

Accuracy for classification model was 96% and 75% on validation data. This was achieved by use of performing transfer learning where the weights from pre trained models in imagenet greatly aided the performance of the model. Only 10 % of the total data was used for training model for classification due to compute resource limitations.

Segmentation and Classification Tool:

The tool created and powered by Voila can be hosted by a private or public server and is able to utilize the models as described in the procedures above. Users are able to upload images or patients files from any plane (sagittal, coronal, and axial) and perform segmentation and or classification. The performance of the segmentation and classification is dependent on the respective models built. Therefore developers can improve models overtime and update directories hosting models with performant version. In addition Segmentation and Classification tool also consists of analysis reports regarding keywords that are available via separate tabs. Results of Segmentation and Classification returns prediction images with the highest pixel count first, which is an indirect way to find the slice or image dominated by the neoplasm.

1) Four separate tabs were created for dashboard.
    a. Tab 1 contained widgets to analyze patient and was labeled Prediction Analysis.
        i. first column original image, second column mask, third column predicted segment,
    b. Tab2 contained widgets to perform segmentation and classify individual images and was
    c. Tab3 contained widgets to view frequency of expert words in documents year over year and was labeled Expert Word Analysis.
    d. Tab4 contained widgets to view n-grams results and was labeled N-Grams

Limitations:

1) Needed additional compute power to build classification model.
2) Images provided for testing should be a mixture of High grade Low grade and No tumor
3) Images for testing should not have any other tissue other than the brain
4) Segmentation model and Classification model are independent of each other and is therefore prone to mismatch results therefore this issue may be alleviated by applying business logic.
5) Analyses is only done on individual brain slices without a holistic representation

Conclusion:

Building a user interface that uses models to segment and predict gliomas from multiple planes will ultimate increase the speed to which patients can be diagnosed as to whether or not they have a glioma without incorporating invasive methods. The tool itself can improve overtime as additional features and preprocessing steps are fine tuned to give more accurate and streamlined results.

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